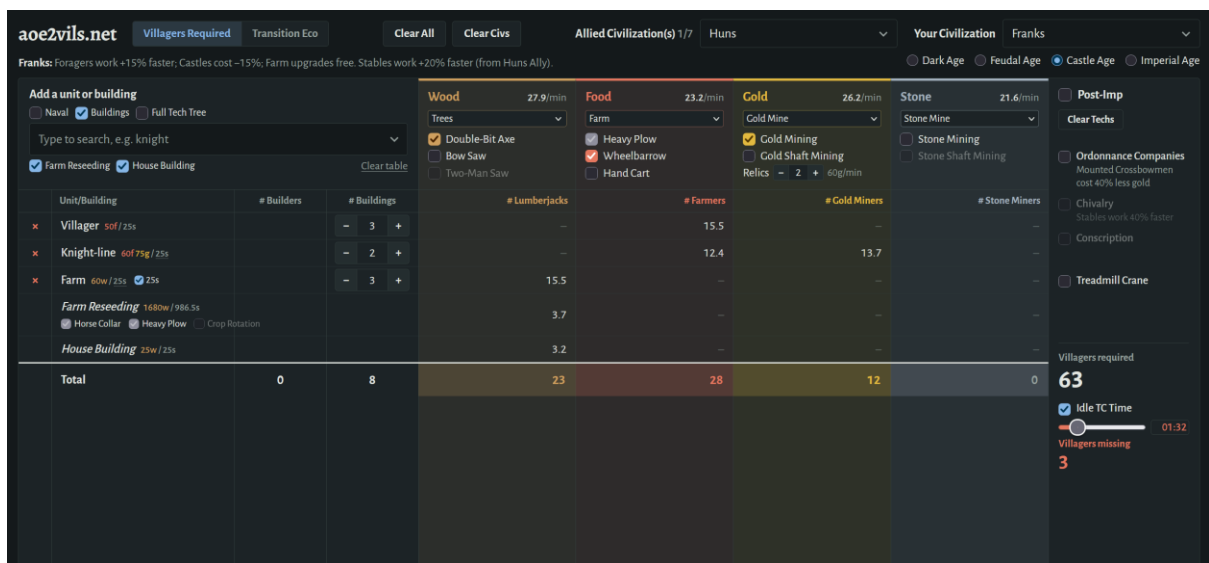


aoe2vils.net overview

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3 October 2026

no LLM nor AI was used in the authoring, creation, editing, design, or crafting of this document.
any and all great and/or dumb things about this document are 100% the responsibility of a human.



Screenshot of the whole website.

Acknowledgments

- [Original villagers required webapp](#) by [Survivalist](#), his [repository](#) was used for the development of aoe2vils.net, with his express permission.
- [aoe2techtree.net](#) by [HSZemi](#) and [Siege Engineers](#), their [repository](#) was used for development of aoe2vils.net, permissible under [MIT license](#).
- The code base of aoe2vils.net was generated by [Claude Opus 5.5](#) with max or extra effort. All prompts and development timeline are publicly available (see context markdown files included in [the github repository](#) or go to [aoe2vils.net/context_a.md](#) and [aoe2vils.net/context_b.md](#))
- Age of Empires II: Definitive Edition © Microsoft Corporation. aoe2vils.net was created under Microsoft's "[Game Content Usage Rules](#)" using assets from Age of Empires II: Definitive Edition, and it is not endorsed by or affiliated with Microsoft.

UI decisions

- Everything on one page (no scrolling)
 - Idea behind: user spends as little time on the site as possible (focus is the user finding out what they want to know as quickly as possible, not spending time interacting with the website)
- No images or tile selections
 - Idea behind: the UI doesn't grow with the game. Minimising how much the user is overwhelmed from the growing size of the game, fast load time, less focus on the site, more focus on accomplishing the user's task with the tool as fast and as easily as possible
- Only one income type viewable at a time
 - Idea behind: less overwhelming, more readable and overviewable, easier control of upgrades

Relics and niche incomes thought of as applied bonuses (as opposed to standalone income types)

- Relics and niche income types subtract from required number of villagers instead of being a standalone income types because:
 - Relics don't make sense to show as standalone income type because relics aren't units assignable to other resources nor an easily obtainable quantity – most of the time there are only 5 relics total on the map.
 - Niche incomes like food from Shu lumberjacks and gold from Varangian fishing are so tiny rates that they don't make useful information

Shu: Lumberjacks generate food in addition to wood.

				Wood 27.9/min	Food 20.3/min	Gold 22.8/min	Stone 21.6/min	Post-Imp
Add a unit or building								
☐ Naval ☒ Buildings ☐ Full Tech Tree								
Type to search, e.g. knight								
☒ Farm Reseeding ☒ House Building								
Clear table								
Unit/Building	# Builders	# Buildings		# Lumberjacks	# Farmers	# Gold Miners	# Stone Miners	
x Villager 50f / 25s		- 1 +			5.9			
x Archer Upgrade 25w 45g / 35s		- 1 +		1.5		3.4		
x Farm 60w / 25s 25s		- 1 +		5.2				
Farm Reseeding 380w / 532.2s				1.5				
☐ Horse Collar ☐ Heavy Plow ☐ Crop Rotation								
House Building 25w / 72.9s				1.1				
Total	0	3		10	5	4	0	19
				4.97 villagers, after 10 lumberjacks cover 0.94				
				Villagers required				
				☐ Idle TC Time				

That 75 Shu lumberjacks per TC sustain villager production is not useful information. Instead, more useful information is that with Shu, having 10 lumberjacks makes you need 5 on food instead of 6 on food to sustain villager production per TC.

Farm reseeding/house building

- Useful way of showing how upgrades are important: see how every additional or missing upgrade from mill and lumbercamp affect needed wood villagers for arbitrary amount of farms
- Easy way of seeing how much quicker you have to start building houses with more production
- Togglable if the user doesn't want to see it or count in the current villager total

Use case example

- While reviewing a recorded game, you want to evaluate the current villager configuration:
 1. Instead of browsing or searching units from all 56 civs, filter relevance of options automatically with 2 steps by selecting civ and age
 2. Starting point: what's the villager total?
 3. What productions can that villager budget afford?
 4. What eco upgrades are missing? How does that affect the needed villager amount?
 5. How many if any villagers have been lost to idle TC time?
 6. Which resource has surplus villagers? -> How the economy could have been improved at this point of the game and how the player can improve this part of the game in future matches

Transition eco

- A tab accessible from the top of the page:
 - Get achieved resources for any age-time (or user-defined time) with any villager configuration
 - Vice versa: get needed villager configuration for any resource goals in any time.

Other benefits of aoe2vils.net which were important to me to ensure

- Easy short and memorable website and domain name
- Consistently less than half a second of load time (only 2 http requests and ~170 kB network transfer)
- No ads, no cookies, no popups
- "future proofing" workaround: user-defined custom units and gather rates

Miscellaneous

- aoe2vils.net is a monolith html file – this means you are able to save the page and run it offline normally
- Buildings have the builders formula ($3t/(n+2)$) programmed so you can see the exact construction time of any building with arbitrary amount of villagers building it